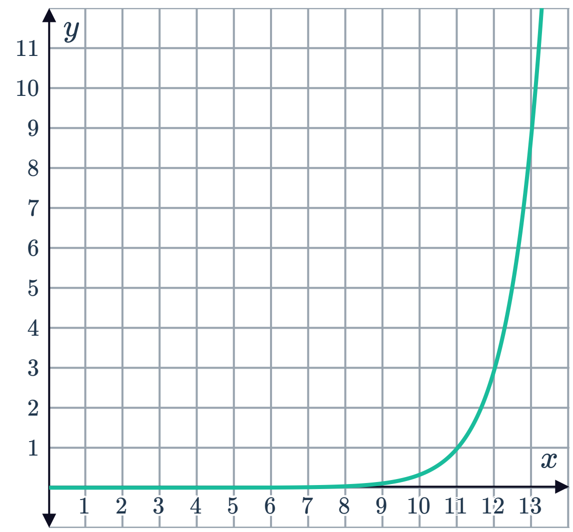
# Exponential regression

**1** The following exponential curve of best fit was used to model the set of data points: $(8, 0.02), (9, 0.1), (10, 0.3), (11.5, 2), (13.5, 15)$



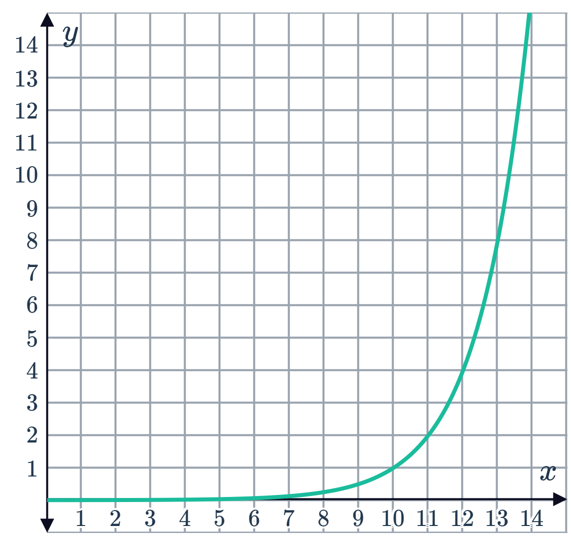

a Predict the $y$-value of a point with an $x$-value of $12$.

b Determine whether the following points would be predicted by the exponential curve of best fit:

i $(13, 9)$  ii $(13, 11)$

iii $(12, 2)$  iv $(11, 2)$

**2** The following exponential curve of best fit was used to model the set of data points: $(7, 0.02), (8, 0.1), (9, 0.5), (10, 1), (11, 2)$




a Predict the $y$-value of a point with an $x$-value of $12$.

b Determine whether the following points would be predicted by the exponential curve of best fit:

i $(6, 0)$  ii $(13, 8)$

iii $(14, 14)$  iv $(12, 6)$

**3** Consider the exponential functions $P$ given by $y = 9\left(3^{-x}\right)$ and $Q$ given by $y = 5\left(3^{-x}\right)$.

a Are the exponential functions $P$ and $Q$ increasing or decreasing?

b Sketch the two functions on the same set of axes.

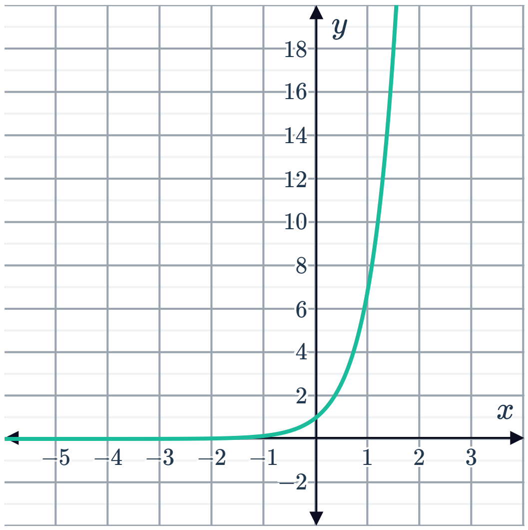
c As $x$ tends to $\infty$, what value does each function approach?

d Describe a transformation of the graph $y = 3^{-x}$ that would obtain:

i $P$  ii $Q$

**4** Consider the following pair of functions:

- Function $P$




- Function $Q$
  $y = 2^x + 4$

**a** Determine how many times the graphs of $P$ and $Q$ intersect.

**b** Determine which function tends to infinity the quickest as $x \to \infty$.

**5** The exponential function $E$ is given by $y = 3^x + 4$ and the exponential function $F$ is given by $y = -8^x + 1$.

**a** Graph the exponential functions on the same coordinate plane.

**b** Identify how many times $E$ and $F$ intersect.

**c** Describe the end behavior of each exponential function as $x \to \infty$ .

## Applications

**6** An independent organization conducts quality testing of various products. One of their aims is to determine if more highly priced goods reflect better quality products. The following scatter plot shows price and quality rating given to 12 different cooking pans on the market.


**a** A new pan comes into the market priced at $\$75$.
Using data from the scatter plot, can the new pan's quality rating be approximated by fitting a function to the data?

**b** True or False: According to the information given, the new pan’s quality rating can be approximated by finding a point on the scatter plot which corresponds to a similar price.

**7** Imran made some syrup and then put it in the fridge to cool down. He measured the temperature of the syrup at $2$-minute intervals for $10$ minutes. The temperature each time is represented in the table.

| Minutes | 0 | 2 | 4 | 6 | 8 | 10 | 12 | 14 |
|---|---|---|---|---|---|---|---|---|
| Temperature ($^\circ$F) | 170 | 143 | 124 | 113 | 105 | 94 | 87 | 85 |

**a** Using technology, sketch a graph showing these values and use it to determine if the data suggests an exponential association. Explain your answer.

**b** Using technology determine an appropriate equation to model the data set to four decimal places.

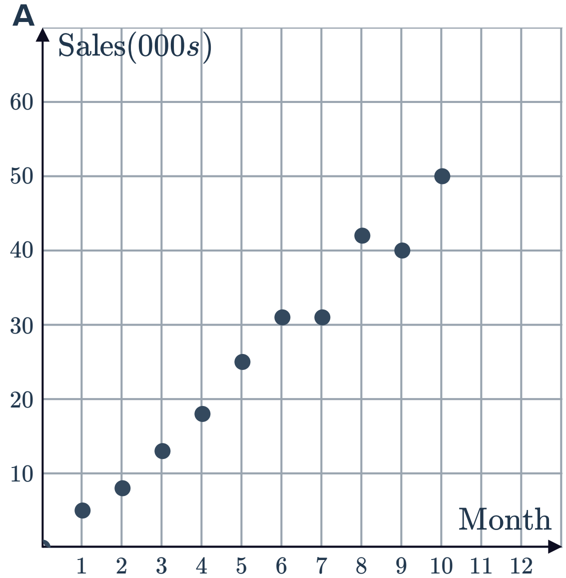
**c** Explain why the function is suitable to model the relationship.

**d** After $20$ minutes, the syrup has cooled enough to eat. Find the temperature that the syrup is cool enough to eat. Round your answer to one decimal place.

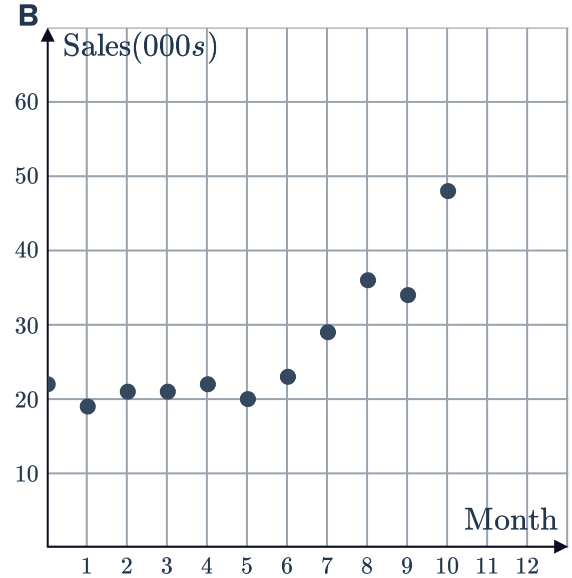
**8** When CTech first released a digital applicati[redacted]n app) onto the market, the number of sales increased slowly at first, but then the number of sales started to increase very rapidly.

**a** Which scatter plot shows the trend in sales over time from when the app was first released?

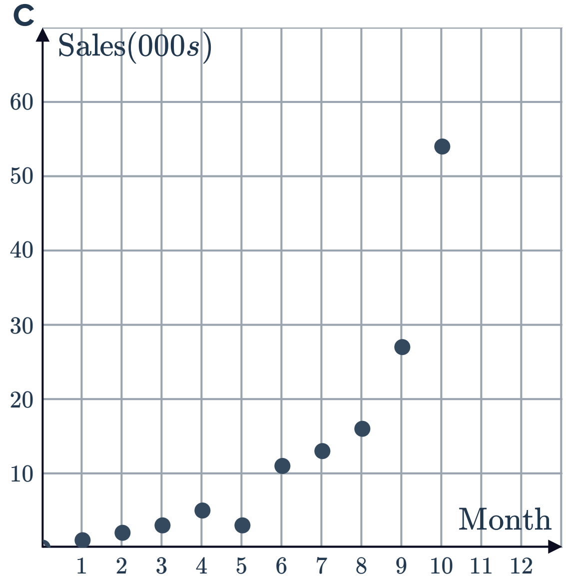
**A**




**B**




**C**




**b** The function $y = 1000\left(50^{\frac{t}{10}} - 1\right)$ is used to approximate the number of sales after $t$ months, where $y$ represents the number of sales.
Complete the table of values for $y = 1000\left(50^{\frac{t}{10}} - 1\right)$.

| $t$ | 0 | 10 |
|---|---|---|
| $y$ | | |

c 20 months after CTech released their ap[illegible]val company, BTech, released a similar app with improved features. In the month that followed, CTech's sales dropped by $60\,000$ from their previous month's sales. According to the model $y = 1000\left(50^{\frac{t}{10}} - 1\right)$, what were CTech's sales one month after BTech released their new app?

9 Mohamad has just started a new job, where his starting salary is $\$50\,000$ p.a. and is expected to increase by $3.2\%$ each year. Elizabeth has also just started a new job, which has a starting salary of $\$49\,000$ p.a. and is expected to increase at a rate of $6.1\%$ each year.

a Write an equation for $M$, Mohamad's salary in $t$ years time.

b Write an equation for $P$, Elizabeth's salary in $t$ years time.

c How much greater will Elizabeth's salary be in $6$ years' time than Mohamad's?

d Over the long term, whose salary will increase more rapidly?

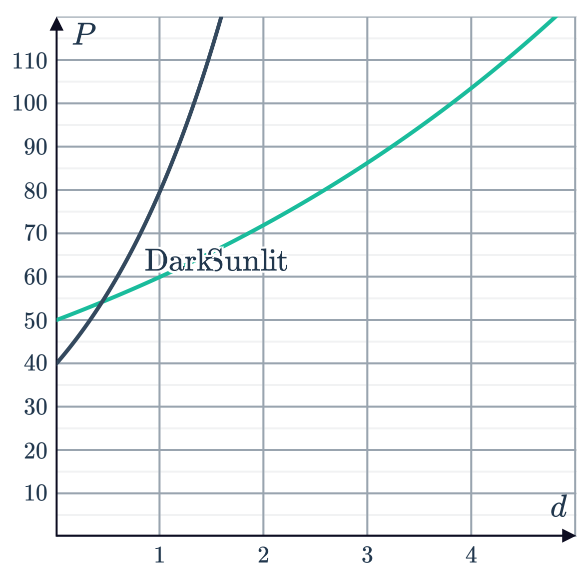
10 Switzerland's population in the next $10$ years is expected to grow approximately according to the model $P = 8\left(1 + r\right)^t$, where $P$ represents the population (in millions) $t$ years from now.
The world population in the next $10$ years is expected to grow approximately according to the model $Q = 7130\left(1.0133\right)^t$, where $Q$ represents the world population (in millions) $t$ years from now.

a According to the model, what is the current population in Switzerland?

b According to the model, what is the current world population?

c Switzerland's population is expected to increase at a slower rate than the world population. Could Switzerland's population increase each year could be $1.79\%$ or $0.87\%$?

11 To investigate the environmental effect on bacterial growth, two colonies of the same bacteria were studied. One was placed in a constantly sunlit environment, while the other in a dark environment. The graph shows the population, $P$, of each colony after a certain number of days, $d$.




a How many more bacteria were initially present in the sunlit environment?

b By what percentage rate did the bacteria in sunlight increase each day?

c By what percentage rate did the bacteria in darkness increase each day?

d Compare the growth rates of the two groups of bacteria

12 Bank A pays interest at a rate of $5.3\%$ p.a. compounded annually, while Bank B offers a rate of $4\%$ p.a. with interest compounded quarterly.

a Form an equation for $A$, the amount $\$P$ would grow to after $t$ years if invested with Bank A.

b Form an equation for $B$, the amount $\$P$ would grow to after $t$ years if invested with Bank B.

c Determine which bank offers the better deal.

13 The median national sale price of property over the next $10$ years can be approximated by the equation $P = 578\,000\,(0.924)^t$, where $P$ represents the median sale price $t$ years from now.
The median national rental price of property over the next $10$ years can be approximated by the equation $Q = 280\,(1.055)^t$, where $Q$ represents the median rental price $t$ years from now.

a According to the models, which will decrease over the next $10$ years, the sale price or rental price of property?

b At what annual percentage rate are rental prices expected to increase over the next $10$ years?

c If interest rates decrease at any time, the annual percentage change in sale prices is expected to increase by $4\%$. If this occurs, what will be the new annual percentage change in sale prices?

14 Kenneth currently works for a company where he sees $90$ clients and the number of clients increases by approximately $5$ each month. He is considering opening his own business which he knows his $90$ clients will follow him to. He anticipates that this will grow by $4\%$ each month.

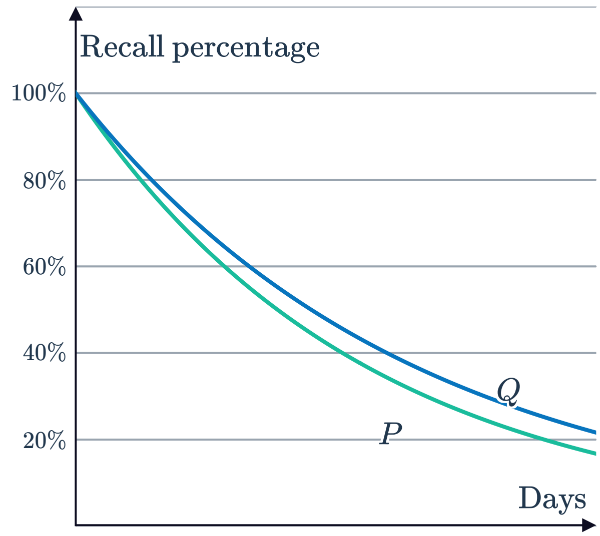
a Form an equation for $P$, the number of clients he will have in $t$ months if he stays at the company.

b Form an equation for $Q$, the number of clients he will have in $t$ months if he opens his own business.

c In which place will he have more clients one year from now?

d In the long term, which option will allow him to increase the number of clients at an increasing rate?

**15** In a memory study, subjects are asked to me[redacted]ze some content and recall as much as they can remember each day thereafter. The given graphs represent the percentage of content remembered by two participants Maximilian ($P$) and Lucy ($Q$) after $t$ days:




**a** Each day, Maximilian finds that he has forgotten $15\%$ of what he could recount the day before. Form a model for the percentage $P$ that Maximilian can still recall after $t$ days.

**b** Who was forgetting more rapidly?

**c** Which one of the following is the model for the percentage that Lucy can still recall after $t$ days?

**A** $Q = (0.85)^t$ **B** $Q = (0.83)^t$

**C** $Q = (0.87)^t$ **D** $Q = (0.81)^t$

**d** On each day, what percentage of the previous day's content did Lucy forget?

**e** According to the models, will either of them completely forget the content?

**16** The population of a country is currently estimated at $2\,410\,000$ and is expected to grow by $2.2\%$ each year. The country’s food supply is currently enough for a population of $3\,374\,000$ people and is set to increase by $2\%$ each year.

**a** Form an equation for $P$, the country’s expected population in $n$ years.

**b** Form an equation for $F$, the number of people the food supply will feed in $n$ years.

**c** According to the models, will there ever be a food shortage?

**d** According to the information, how do we know that there will eventually be a food shortage?

**e** When will the food shortage occur?

**17** Two people are asked to eat the same food [redacted]ual quantity. They were chosen such that Person A incorporates lots of sugar into their diet while person B consumes limited amounts of sugar in their diet.
After eating, their blood sugar level (amount of sugar in the bloodstream) peaked, and was then continually measured, with the following functions modeling their blood sugar over time:

- Person A: $f(t) = 179(0.3)^t$, where $f(t)$ represents blood sugar level at time $t$ hours
- Person B: $g(t) = 132(0.6)^t$, where $g(t)$ represents blood sugar level at time $t$ hours

**a** At what value did Person A's blood sugar level peak?

**b** At what value did Person B's blood sugar level peak?

**c** Which person's blood sugar level decreased more gradually?

**d** By what factor did Person A's blood sugar level decrease each hour?

**e** From Person A and Person B's results, what can be concluded?

**18** The population of two different bacteria, labeled $J$ and $K$, are given by the tables below:
Bacteria $J$:

| $t$ (time in days) | $0$ | $1$ | $2$ | $3$ | $4$ |
|---|---|---|---|---|---|
| $P$ (population) | $1$ | $60$ | $3\,600$ | $2.16 \times 10^5$ | $1.296 \times 10^7$ |

Bacteria $K$:

| $t$ (time in days) | $0$ | $1$ | $2$ | $3$ | $4$ |
|---|---|---|---|---|---|
| $Q$ (population) | $1$ | $50$ | $2\,500$ | $1.25 \times 10^5$ | $6.25 \times 10^6$ |

**a** Does the population of bacteria $J$ increase slower or faster than the population of $K$?

**b** The population $P$ of bacteria $J$ at time $t$ can be modeled using the general equation $P(t) = a^t$, where $a > 1$. By using the table of values, graph $P(t)$ for $t > 0$.

**c** The population $Q$ of bacteria $K$ at time $t$ can be modeled using the general equation $Q(t) = b^t$, where $b > 1$. By using the table of values, graph $Q(t)$ for $t > 0$.

**d** Can the population of bacteria $K$ be found by scaling the population of bacteria $J$ by a factor?